INTELLIFY HACKATHON 3.0

Introduction:

Intellify is a National-level Hackathon designed to foster innovation and collaboration in the fields of Software and Hardware. Hosted by Marwadi University(Dept. Of AI, ML, DS), this event aims to bring together creative minds from across India to solve real-world problems through cutting-edge technology.

Guidelines:

Open to all UG, PG, and Research scholars from any accredited institute across India.

- Inter-college teams are not allowed.
- Inter-specialization teams are allowed.
- The Hackathon consists of 3 rounds:

    1. Round 1 & Round 2: Online
    2. Round 3 (Grand Finale): Offline

Rules:

- Teams must have a minimum of 3 and a maximum of 5 participants.
- Attendance in the first two rounds is mandatory for final selection.
- Registration fee: Rs. 500 (INR) per team.

Evaluation Criteria:

Projects will be evaluated based on the following criteria:

- Judging Criteria
- Projects will be evaluated based on:
- Creativity & Innovation
- Technical Complexity
- Impact & Relevance
- Feasibility & Implementation

PROBLEM STATEMENTS

Track: Software

IH3MU01: Social Media Analyzer(By TSS)

1. Overview:

The Social Media Analyzer is an AI-powered system designed to ethically extract publicly available data from social media platforms to create comprehensive user profiles. This tool helps analyze online activity to assess sentiment, reputation, and potential risks while ensuring data privacy and security.

2. Objectives:

- Automate the collection of publicly available user data from social media.
- Structure and analyze the collected data to create user profiles.
- Identify patterns that indicate positive and negative attributes based on user activity.
- Ensure ethical and legal compliance in data handling.

3. Key Features:

3.1 Data Collection

- Extracts publicly available data (e.g., name, location, connections, interests) from social media platforms.
- Avoids collecting private or restricted data to maintain ethical compliance.

3.2 Profile Creation

- Structures the collected data into a user profile using NLP techniques.
- Stores structured profiles in a database for further analysis.

3.3 Sentiment & Risk Analysis

- Identifies positive indicators such as awards, achievements, and recognition.
- Flags potential negative content such as extremist views or suspicious activity.
- Uses keyword analysis, image detection, and behavioral patterns for assessment.

4. Input Data:

Users can provide one of the following input options:

4.1 Broad Search (Option 1: Name, Mobile Number, Email)

If the user provides general identifiers such as:

- Full Name
- Mobile Number
- Email Address

A broad search is conducted across the internet, including:

- Public databases
- Directories
- Professional networks
- Social media platforms where such details are publicly available

4.2 Narrow Search (Option 2: Social Media IDs)

If the user provides specific social media handles or profile URLs, a targeted search is performed directly on those platforms. This ensures that only data from the specified social media sources is retrieved. Supported inputs include:

- Facebook ID
- Instagram Handle
- LinkedIn Profile (or URL)
- Twitter Handle
- YouTube Channel URL

5. Expected Output:

- **Part A**: Extracted images and publicly available data, including user details, connections, and engagement metrics.
- **Part B**: Reputation and sentiment analysis, highlighting both positive indicators (awards, achievements, professional credibility) and negative indicators (controversial content, flagged behavior, suspicious activity patterns).

6. Conclusion:

The Social Media Analyzer helps organizations and individuals assess online presence and sentiment based on publicly available data. By maintaining ethical standards and focusing on user behavior analysis, it provides insights into online reputation and risk factors while ensuring ethical and responsible profiling.

IH3MU02: Court Order Information Extraction and Summarization(By TSS)

1. Overview:

The Court Order Information Extraction and Summarization project aims to streamline legal data processing by developing an AI-powered system capable of extracting structured information from complex legal documents and generating concise summaries.

2. Objective:

The primary objective of this project is to develop a robust and scalable tool that can:

- Extract key information from court orders in a structured JSON format, including details such as case number, order date, judge name, party information, designations, addresses, age, aliases, and other relevant metadata.
- Generate concise summaries that highlight the most critical elements of the judgment.
- Ensure high performance.

3. Scope:

- **Information Extraction:** Development of algorithms and techniques to accurately identify and extract pertinent information in a json format of the following details-

Case Number
- Order Date
- Name of Judge
- List of Parties
    - Type of Party (individual, government, NGO, etc)
    - Name of party
    - Official designation
    - Address (apartment, street, village, city, district, state, zipcode)
    - Age
    - Other mentioned detail (Alias, Other Address, Relations)

- **Summarization:** Creation of a system to generate clear and concise summary

that capture the essential elements of a court case in 100-120 words.
- It should cover details from the final judgement including fines, penalties, prison sentences along with their exact number.
- It should mention any contingencies mentioned in the judgement.
- It can additionally cover any directions issued or any acquittals.

4. Expected Outcome:

- **Performance Metrics:**

  The code should be able to run fulfilling the given criteria

  - Process one page in under 10 seconds.
  - Handle documents up to 100MB and 400 pages.
  - Operate fully offline once setup

5. Resources:

Participants will have access to the following resources:

1. 10 Solved Sample Court Orders
   a. Court Order PDF Document.
   b. Extracted details available in JSON format.
   c. A text file summarizing the court order.
2. 20 additional court orders in PDF format for testing the solution.
3. Technical support to address participant queries related to the problem statement.

6.Conclusion:

This project has the potential to significantly improve the efficiency and effectiveness of handling legal documents by automating the extraction and summarization process. By providing structured outputs and clear summaries, the tool can assist legal practitioners, analysts, and institutions in reducing manual effort, minimizing errors, and accelerating legal research and decision-making.

IH3MU03: Healthcare Analytics

1. Overview:

Develop a predictive model using machine learning algorithms that can predict the likelihood of a patient developing a certain medical condition. The model should be trained on a large dataset of patient medical records and should be able to identify risk factors and provide recommendations for preventative care.

2. Aim:

To utilize data analytics techniques to uncover patterns, improve patient care, reduce operational costs, and enhance decision-making in healthcare services.

3. Objective:

- To collect and preprocess patient health records and hospital management data for early disease prediction.
- To apply descriptive and predictive analytics for identifying trends and forecasting health risks.
- To evaluate hospital performance metrics like patient wait times, readmission rates, and treatment effectiveness.
- To optimize resource allocation(staff, equipment) using historical data.
- To detect fraud in insurance claims using anomaly detection

4. Expected Output:

- Summary statistics on disease prevalence, treatment outcomes, and patient demographics.
- Predictive models for disease risk (e.g., diabetes, heart disease).
- Anomaly detection in lab results or patient vitals.
- Visual dashboards for:

    1. Hospital utilization (beds, staff)
    2. Patient flow and readmission patterns
    3. Disease trends over time

5. Conclusion:

Healthcare analytics enables healthcare providers to make informed decisions, optimize operations, and deliver better patient care.This will help improve patient outcomes, anticipates public health trends, efficient resource management and enables proactive healthcare management.

IH3MU04: Data-Driven AI for Sustainable Farming

1. Overview:

Develop an AI system which aims to enhance agricultural productivity through intelligent data analysis and decision-making. This system will help farmers to make informed choices about crop selection, irrigation, fertilization, and pest control, ultimately improving yield.

2. Aim:

To develop an AI-powered system that supports sustainable agriculture by optimizing crop yield, reducing resource wastage, and improving farming decisions through data analysis and intelligent predictions.

3. Objectives:

- To collect and analyze data related to soil health, weather conditions, crop type, and farming practices.
- To optimize irrigation and fertilizer usage using real-time sensor data
- To detect plant diseases early using image recognition
- To assist in precise irrigation, fertilization, and pest control using machine learning and IoT integration.
- To enable farmers to make data-driven decisions like recommending crop rotation strategies for soil sustainability
- To provide an easy-to-use dashboard or mobile interface for real-time monitoring and recommendations.

4. Expected Output:

- AI-based recommendations for optimal crop selection based on soil and climate conditions.
- Predictive alerts for irrigation scheduling and rainfall.
- Fertilizer and pesticide usage plans tailored to current soil and crop conditions.
- Early detection of crop diseases or pest infestations.
- Yield forecasts and resource usage reports.
- Interactive dashboards for real-time soil and crop health monitoring.

5. Conclusion:

The Data-Driven AI system for sustainable farming empowers farmers by boosting productivity while minimizing resource waste, thereby reducing environmental harm. By leveraging AI and data science, the system supports individual decision-making and provides real-time actionable insights.

IH3MU05: Optimizing Retail Inventory with Multi Agents

1. Overview:

Develop a system that aims to improve stock management, reduce overstock and stockouts, and enhance decision-making in retail through intelligent agent-based systems. The system continuously learns and adapts, helping retailers respond quickly to market changes, reduce waste, and improve customer satisfaction.

2. Aim:

To develop an intelligent multi-agent system that automates and optimizes retail inventory management by minimizing overstock and stockouts, improving responsiveness to market demands, and enhancing overall operational efficiency.

3. Objectives:

- To design and implement a distributed agent-based system for real-time inventory monitoring and decision-making.
- To enable collaboration among intelligent agents for tasks like demand forecasting, restocking, and supplier coordination.
- Minimize perishable/seasonal item spoilage to cut waste
- To enhance responsiveness to sales trends and customer preferences using machine learning and adaptive strategies.
- Optimize pricing, discounts, and warehouse allocation in order to improve profit margins

4. Expected Output:

- Dynamic restocking recommendations based on demand prediction
- Alerts for low stock or overstock situations
- Simulation reports showing effects of different stocking strategies
- Real-time dashboards for inventory status and agent decisions

5. Conclusion:

The multi-agent inventory optimization system transforms traditional stock management by introducing intelligent, decentralized decision-making. Through collaboration, adaptability, and predictive analytics, it ensures the right products are available at the right time and place, thereby reducing waste, cutting costs, and elevating the customer shopping experience.

IH3MU06: AR/VR Visualizations for Interior Designers

1. Overview:

An immersive platform that lets designers and clients to visualize and interact with interior designs in real-time 3D environments. Using AR/VR technology, it enhances design accuracy, client engagement, and decision-making.

2. Aim:

To develop an immersive AR/VR-based platform that allows interior designers and clients to visualize, explore, and customize interior designs in real-time 3D environments, enhancing collaboration, design accuracy, and decision-making.

3. Objectives:

- To enable 3D visualization with real-time lighting/texture adjustments.

- To allow users to interactively place, move, and modify furniture, decor, and layout within the virtual environment.
- To reduce the time and cost of redesign by enabling instant visualization and feedback.
- Swap design elements instantly during client meetings.

4. Expected Output:

- Fully navigable and interactive 3D models of interior spaces
- AR mode for overlaying virtual designs on real-world environments
- VR walkthroughs for immersive experience using headsets
- Customization tools for changing furniture, colors, and layouts in real-time
- Rendered design snapshots or video previews for client presentation

5. Conclusion:

The AR/VR interior design visualization system revolutionizes traditional design workflows by providing immersive, interactive experiences. It enables designers and clients to collaboratively explore and modify virtual spaces in real time, it eliminates costly redesigns, and speeds up decision-making.

IH3MU07: Decentralized Supply Chain Tracking System

1. Overview:

Develop a blockchain-based system to track goods through a supply chain, ensuring transparency, authenticity, and accountability. Focus on a specific industry (e.g., food, pharmaceuticals) and demonstrate the benefits of decentralization.

2. Aim:

To design and develop a decentralized software platform that tracks the journey of goods through a supply chain using blockchain technology, thereby enhancing transparency, traceability, and trust among all stakeholders.

3.Objectives:

- Implement smart contracts to record and verify transactions between supply chain actors.
- Develop a web/mobile interface for users to interact with the blockchain, view product history, and update statuses.
- Ensure data immutability by storing critical records (e.g., ownership transfer, timestamps, quality checks) on-chain.
- Enable product verification by customers using batch IDs or QR codes.

- Demonstrate industry application (e.g., food or pharmaceutical) with test data and simulated workflow.

4.Expected Outcome:

- A fully functional decentralized web application that can:
- Record product movement across a supply chain via blockchain
- Display a tamper-proof audit trail of each product
- Allow role-based access for producers, suppliers, and consumers
- Detect unauthorized modifications or gaps in the supply chain
- Increased trust and data integrity in supply chain management

5.Conclusion:

This project showcases how cutting-edge blockchain technology can revolutionize supply chain systems by eliminating fraud, enhancing visibility, and improving efficiency. By decentralizing data storage and using smart contracts to automate processes, the system promotes accountability and trust, especially in industries where product authenticity and quality are critical.

IH3MU08: Maternal Health Monitoring App for Women's

1. Overview:

The Maternal Health Monitoring App helps women track their pregnancy health by monitoring vital signs, providing personalized advice, reminders, and alerts. The system collects data from users and wearable devices and offers tailored advice based on medical guidelines, supporting timely care.

2. Aim:

To develop a mobile application that supports maternal health by continuously monitoring vital signs, providing personalized guidance, and ensuring timely care for expectant mothers through real-time data and smart alerts.

3. Objectives:

- To track essential maternal health indicators such as heart rate, blood pressure, fetal movements, glucose levels, and weight.
- To provide personalized advice and flag abnormalities based on pregnancy stage and medical history.
- To send automated reminders for medication, doctor visits, and health checks.
- To integrate with wearable health devices for real-time monitoring.

- To alert users and caregivers in case of abnormal vital readings or high-risk conditions.

4. Expected Output:

- Real time health dashboards with trend analysis.
- Personalized health tips and nutritional advice
- Alerts for abnormal vitals or symptoms (e.g., high blood pressure, gestational diabetes)
- Reminder notifications for medication, hydration, and appointments

5. Conclusion:

The Maternal Health Monitoring App empowers women with the tools to manage their pregnancy proactively. By combining health tracking, personalized insights, and timely alerts, the app promotes safer pregnancies, early risk detection and improved maternal outcomes.

IH3MU09: Crowdsourced Pothole & Road Hazard System

1. Overview:

This project aims to empower citizens to participate in road safety and maintenance by enabling real-time reporting of such issues. Using a mobile or web platform, users can upload photos, GPS locations, and descriptions of hazards, which are then routed to concerned municipal authorities for quick resolution.

2. Aim:

Develop a reporting system that aims to improve road safety and maintenance by enabling citizens to report potholes and hazards in real-time with locations, photos and description and alerts authorities for prompt actions.

3. Objectives:

- Create a user-friendly app/web interface for citizens to report issues using images, text, and live GPS.
- Integrate geolocation and mapping (e.g., Google Maps) to pin the exact location.
- Automate alerts to notify relevant local authorities with status tracking.
- Build an admin dashboard for authorities to manage reports, update statuses, and view heatmaps.

IH3MU10: AI-Driven Optimization of Urban Traffic Flow for Reduced Emissions

1. Overview:

An AI system that analyzes real-time traffic data and optimizes traffic light timings and routing strategies to minimize traffic congestion and reduce vehicle emissions in urban areas.

2. Aim:

To develop an AI-based system that utilizes real-time traffic data to optimize traffic signal timings and routing strategies, thereby reducing urban traffic congestion and lowering vehicular emissions.

3. Objectives:

- To analyze real-time traffic data collected from sensors, GPS, and cameras across urban intersections.
- To predict traffic patterns using machine learning models, particularly time-series forecasting.
- To implement reinforcement learning algorithms that dynamically adjust traffic signal timings.
- To simulate various routing and signal timing strategies using traffic simulation tools.
- To measure and minimize carbon emissions through optimized traffic flow and reduced idle time.

4. Expected Output:

- Adaptive traffic light control strategies for individual intersections and entire road networks
- Optimized routing suggestions to reduce bottlenecks and travel time
- Quantified reductions in carbon emissions and fuel consumption
- Visualization dashboards showing traffic flow, heatmaps, and environmental impact

5. Conclusion:

The AI-driven urban traffic optimization system leverages cutting-edge machine learning techniques and real-time data processing to make cities more efficient and environmentally friendly. By reducing traffic congestion and emissions, it supports sustainable urban development and enhances the quality of life for city dwellers. The solution also contributes to global efforts in reducing greenhouse gas emissions and combating climate change.

IH3MU11: AI-Based Sign Language Translator

1. Overview:

This project aims to bridge that gap using AI-based computer vision and natural language processing to develop a real-time sign language translator.

2. Aim:

To design and implement an AI-based system that recognizes sign language gestures and translates them into text or speech in real time, enabling seamless communication between signers and non-signers.

3. Objectives:

- Gather or use publicly available sign language datasets .
- Use computer vision and deep learning (e.g., CNNs or Transformers) to detect and classify hand gestures from video or camera input.
- Convert recognized signs into grammatically correct text or speech using NLP techniques.
- Develop a real-time interface for signers and non-signers to interact.
- Integrate text-to-speech (TTS) engine to vocalize translated output.

4. Expected Output:

- Real-time display of recognized words or sentences in text format or audio format.
- Synthetic signing animations for reverse translation.
- Interactive tutorials for learning sign language.
- Visual feedback on detected gesture and confidence score

5. Conclusion:

The AI-Based Sign Language Translator bridges the communication gap between the deaf/mute community and the hearing world. By leveraging computer vision and deep learning, it delivers real-time, accurate translation of sign language into speech or text, fostering inclusivity, accessibility, and independence. This technology holds the potential to revolutionize interactions in diverse settings—from classrooms to hospitals—by making communication barrier-free.

IH3MU12: Seamless Event Management System

1. Overview:

Develop a system that aims to simplify and automate the entire event lifecycle—from planning and registration to execution and feedback. The system should enable organizers to manage events effortlessly through a centralized platform, automate repetitive tasks, track participant engagement, and provide analytics for future improvement.

2. Aim:

To develop an integrated platform that streamlines the entire event lifecycle—from planning and registration to execution and feedback—by automating tasks, enhancing coordination, and providing real-time insights for better event experiences.

3. Objectives:

- To create a centralized system for managing all types of events including registrations, ticketing, budgeting, resource allocation and scheduling.
- Provide interactive features like live polls, Q/A for participants.
- To gather post-event feedback and generate analytics for continuous improvement.

4. Expected Output:

- Event dashboards with live updates on attendance, schedule, and logistics
- Automated invitations, reminders, and follow-ups
- Real-time notifications and alerts for organizers and participants
- Feedback reports, engagement analytics, and success metrics
- Task management tools for the organizing team

5. Conclusion:

The Seamless Event Management System transforms traditional event planning into an efficient, automated, and user-friendly experience. It empowers organizers with tools to manage complex logistics, track real-time engagement, and collect actionable insights, resulting in more successful and impactful events.

IH3MU13: Smart Inventory Management System with QR Code Access

1. Overview:

This project proposes a smart, automated inventory system that uses QR codes to track stock movements, reduce human error, and optimize resource utilization. It enables users to scan QR codes for quick access to item details, and allows managers to monitor inventory levels, generate reports, and receive alerts for low stock—all in real time via a centralized dashboard.

2. Aim:

Develop a model such that it aims to create an efficient and accurate way to manage inventory by replacing manual tracking with a QR code-based digital solution. The system should allow users to easily add, update, locate, and manage stock items in real time.

3. Objectives:

- Generate and assign QR codes to each inventory item, containing unique metadata (name, type, date, status, etc.).
- Develop a mobile/web app to scan QR codes and retrieve or update item details.
- Build a centralized dashboard to view, sort, and manage stock items.
- Track inventory status (available, checked out, under maintenance, low stock).
- Send notifications for critical actions like low inventory or expiry dates.
- Enable role-based access control for staff, admins, and auditors.
- Export automated reports for audit or stock review.

4. Expected Outcomes:

- QR code generation and scanning
- Real-time inventory tracking and editing
- Secure user access and logging
- Analytics and summary report.
- Improved stock visibility and faster inventory operations

5. Conclusion:

This system transforms traditional inventory tracking into a modern, automated, and accurate process using QR code technology. It offers real-time data access, reduces overheads, and enhances transparency—making it ideal for organizations looking to digitally upgrade their inventory practices.

IH3MU14: AI Risk Mitigation System

1. Overview:

The increasing reliance on AI systems brings risks such as bias, hallucinations, privacy violations, and adversarial attacks. These risks can lead to ethical, legal, and security concerns. The AI-Risk Mitigation System is designed to automatically identify, evaluate, and reduce these risks to make AI applications more trustworthy, fair, and safe.

2. Aim:

Build an AI system that identifies, evaluates, and mitigates risks in AI systems—such as bias, privacy violations, hallucinations, or adversarial vulnerabilities.

3. Objectives:

- Detect bias, hallucinations, PII leakage, and adversarial inputs in real-time.
- Assign severity scores using standard metrics and a risk matrix.
- Apply appropriate countermeasures to reduce or eliminate the identified risks.
- Generate reports and logs for audit and compliance purposes.
- Ensure the system can be easily integrated into existing AI workflows.

4. Expected Outcomes:

- A working prototype that monitors inputs/outputs of AI systems.
- Detection modules for: Fairness/Bias detection, Hallucination checking, Privacy violation detection, Adversarial input resistance
- A risk scoring engine to prioritize actions.
- A mitigation module to automatically handle or suggest fixes.
- A dashboard for visual insights and alerts.

5. Conclusion:

The AI-Risk Mitigation System addresses the urgent need for responsible AI by proactively identifying and resolving critical issues in model behavior. By embedding fairness, security, and privacy checks into the AI lifecycle, this system promotes safer deployment of AI technologies in real-world applications.

IH3MU15: Virtual Herbal Garden

1. Overview:

This project aims to build a digital platform that provides users with an interactive and immersive experience to explore, learn, and understand a wide variety of medicinal plants used across AYUSH systems.

2. Aim:

Create a Virtual Herbal Garden that provides an interactive, educational, and immersive experience to users, showcasing the diverse range of medicinal plants used in AYUSH (Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy).

3. Objectives:

- Digitally model or visualize a wide range of Indian medicinal plants with descriptions, 3D views, or AR/VR elements.
- Provide scientific and Ayurvedic data for each herb—botanical name, usage, health benefits, region, and preparation.
- Implement interactive features like plant search, audio narration, quizzes, and multi-language support.
- Include filters by disease/benefit (e.g., for immunity, digestion, skin).
- Enable a virtual garden tour with narration or guided learning.
- Promote conservation awareness of endangered herbal species.

4. Expected Output:

- A functional web/mobile app simulating a herbal garden experience.
- An informative herbal plant library with pictures and audio/video content.
- Increased awareness of herbal remedies and sustainable gardening.
- A personalized virtual gardening experience to promote learning by engagement.

5. Conclusion:

The Virtual Herbal Garden offers a unique blend of technology and tradition, helping users reconnect with nature through digital means. It promotes environmental awareness, health literacy, and self-reliance in herbal care.

Track: Hardware

IH3MU16: DIY Prosthetic Hand/Arm with Basic Functionality

1. Overview:

Design and build a low-cost, open-source prosthetic hand or arm using readily available materials and components (3D printing, microcontrollers, actuators). Focus on achieving basic grasping and manipulation tasks.

2. Aim:

To design and develop a low-cost, open-source prosthetic hand or arm that enables users to perform basic grasping and manipulation tasks using accessible materials and modern fabrication technologies like 3D printing and microcontrollers.

3. Objectives:

- To create an affordable and lightweight prosthetic hand/arm using 3D-printed parts.
- To integrate actuators and sensors to enable basic movements such as open/close, grasp, and rotate.
- To control the prosthetic using microcontrollers (e.g., Arduino) and simple input signals (e.g., button, EMG, or flex sensors).
- To ensure the design is open-source and modifiable for global accessibility.

4. Expected Output:

- A functional prosthetic prototype capable of basic hand/arm movements
- Predefined grasp patterns (e.g., power grasp, pinch grasp)
- User-controllable movement via buttons, muscle signals, or joystick
- Documentation for fabrication, assembly, and programming

5. Conclusion:

The DIY Prosthetic Hand/Arm project aims to empower individuals and communities by offering a practical, affordable solution to limb loss. By leveraging 3D printing, microcontrollers, and open-source design principles, it makes basic prosthetic functionality accessible to a wider audience.

IH3MU17: Remote Patient Monitoring System for Elderly Care

1. Overview:

Develop a non-intrusive hardware system for remote patient monitoring of elderly individuals at home. Focus on vital signs (heart rate, respiration), fall detection, and activity tracking, transmitting data securely to caregivers or healthcare providers.

2. Aim:

To design and develop a non-intrusive, IoT-based hardware system that enables real-time monitoring of elderly individuals at home, focusing on vital signs, fall detection, and activity tracking, with secure data transmission to caregivers and healthcare providers.

3. Objectives:

- To monitor key vital signs including heart rate, respiratory rate, and body temperature.
- To detect falls and unusual inactivity patterns using motion and position sensors.
- To provide real-time alerts and periodic health summaries to caregivers or medical professionals.
- To ensure secure and reliable data transmission using IoT and edge computing technologies.

4. Expected Output:

- Real-time dashboards for caregivers with vital signs and activity levels
- Alerts and notifications in case of abnormal readings or fall detection
- Daily and weekly health and activity reports
- Predictive insights for early detection of health deterioration
- Data logs stored securely in cloud or local healthcare systems

5. Conclusion:

The Remote Patient Monitoring System enhances the safety and well-being of elderly individuals by continuously and non-intrusively tracking their health. Leveraging wearable sensors, IoT communication, and edge computing, it enables timely intervention in emergencies and supports proactive healthcare.

IH3MU18: Smart Traffic Management System for Pedestrian Safety

1. Overview:

Develop a hardware system to improve pedestrian safety at intersections. This could involve smart traffic lights that dynamically adjust timing based on pedestrian presence detected by sensors (cameras, LiDAR), or audible/visual alerts to drivers and pedestrians.

2. Aim:

To develop an intelligent hardware system that enhances pedestrian safety at traffic intersections by dynamically adjusting traffic signals and providing real-time alerts based on pedestrian detection using advanced sensors and edge computing.

3. Objectives:

- To detect pedestrian presence and movement at intersections using cameras and LiDAR sensors.
- To dynamically adjust traffic light timings to prioritize pedestrian crossing while maintaining traffic flow.
- To provide audible and visual alerts to both pedestrians and drivers to enhance awareness.
- To implement real-time data processing and decision-making using edge computing devices.
- To ensure system reliability and low latency in diverse environmental conditions.

4. Expected Output:

- Adaptive traffic light control signals that adjust timing based on pedestrian presence.
- Audible alerts (beeps, voice prompts) and visual signals (flashing lights, countdown timers) at crosswalks.
- Real-time dashboard showing pedestrian and vehicle status to traffic control centers.
- Alerts sent to connected vehicles or mobile apps for enhanced driver awareness.
- Logs of pedestrian crossing events and system performance metrics.

5. Conclusion:

The Smart Traffic Management System leverages cutting-edge sensor technology and real-time processing to create safer intersections for pedestrians. By intelligently adapting traffic signals and providing timely alerts, the system reduces accidents and enhances the overall urban mobility experience. It represents a vital step towards smarter, more inclusive cities prioritizing vulnerable road users.

IH3MU19: Smart Home Energy Management Device

1. Overview:

Design a hardware device that optimizes energy consumption in a home by monitoring appliance usage, renewable energy generation (if any), and grid prices. The device should suggest energy-saving strategies and potentially automate appliance scheduling.

2. Aim:

To design and develop a smart hardware device that monitors home energy consumption, renewable energy generation, and grid electricity prices, optimizing energy usage by providing recommendations and automating appliance control to reduce costs and environmental impact.

3. Objectives:

- To continuously monitor energy consumption of various home appliances using IoT sensors and smart plugs.
- To integrate data from renewable energy sources like solar panels to optimize self-consumption.
- To track real-time grid electricity prices and adapt energy usage accordingly.
- To provide users with actionable energy-saving strategies via a mobile or web app.
- To automate appliance scheduling based on energy availability, cost, and user preferences.
- To ensure secure and reliable communication using home automation protocols.

4. Expected Output:

- Dashboard displaying current and historical energy consumption and generation.
- Real-time alerts and suggestions for reducing energy use or shifting load.
- Automated control signals to appliances (e.g., turning off/on or delaying operation).
- Energy cost savings reports and environmental impact summaries.
- Configurable schedules optimized for cost and energy efficiency.
- analytics and better optimization.

5. Conclusion:

The Smart Home Energy Management Device empowers homeowners to take control of their energy consumption by leveraging real-time monitoring, intelligent automation, and data-driven recommendations. This system not only reduces electricity bills but also supports sustainable living by maximizing the use of

renewable energy and minimizing grid dependence, paving the way for smarter, greener homes.

IH3MU20: Smart Irrigation System with Weather Prediction

1. Overview:

This project automates watering based on soil moisture levels and upcoming weather forecasts to conserve water and maintain optimal soil health.

2. Aim:

To design an automated irrigation system that optimizes water usage by monitoring soil moisture and incorporating weather forecast data, ensuring efficient watering and promoting sustainable soil health.

3. Objectives:

- To continuously monitor soil moisture levels using IoT soil sensors.
- To collect and integrate local weather forecast data to predict rainfall and temperature.
- To automate irrigation schedules based on real-time soil moisture and predicted weather.
- To conserve water by avoiding unnecessary irrigation during rainfall or high moisture conditions.
- To provide users with alerts and control options via a mobile or web interface.
- To support scalability for use in small gardens to large agricultural fields.

4. Expected Output:

- Automated control signals to irrigation valves or pumps
- Real-time dashboard showing soil moisture status and upcoming weather
- Alerts and notifications about irrigation schedules and water usage
- Reports on water savings and soil moisture trends
- Optimized watering cycles that improve crop yield and reduce water waste

5. Conclusion:

The Smart Irrigation System with Weather Prediction provides an intelligent solution to water management by combining soil moisture sensing with weather forecasts. It enables precise irrigation, conserves water, and supports healthier plants and crops. This approach helps farmers and gardeners optimize resources while contributing to environmental sustainability.

IH3MU21:Target Detection By Anomaly Detection

1. Overview:

Target Detection by Anomaly Detection refers to identifying unusual patterns or outliers that signify the presence of a "target" or threat, without needing labeled examples for every possible target. It is especially useful in dynamic environments where normal behavior is predictable but anomalies (targets) are rare and varied.

2. Aim:

Develop a target detection model by optimizing Anomaly Detection in Hyperspectral Image Processing using AI/ML.

3. Objectives:

- Gather normal data from sensors, cameras, or logs.
- Identify relevant features that describe the system's behavior (e.g., movement, shape, size, pixel variance, log patterns).
- Train an anomaly detection algorithm (e.g., Isolation Forest, Autoencoders, One-Class SVM) on the normal data.
- Use the trained model to flag instances that deviate from the norm as potential targets.
- Measure performance using precision, recall, and false positive rates.

4. Expected Outcomes:

- A working anomaly detection model that can detect: Intrusions in surveillance footage, Defects in industrial production lines, Network attacks in cybersecurity logs.
- Improved detection rate of unknown or rare events without needing exhaustive labeled datasets.
- Low false positive and false negative rates.
- A real-time monitoring dashboard showing detected targets.

5. Conclusions:

Target detection through anomaly detection offers a scalable and adaptive solution for identifying rare or evolving threats. Unlike traditional supervised models, it requires minimal labeled data and can generalize to new or unexpected targets. This makes it particularly suitable for dynamic, high-risk environments such as security systems, healthcare monitoring, or autonomous vehicles.

IH3MU22: Wearable Sensors for Prevention of Falls in Elderly People

1. Overview:

Wearable sensors provide a non-intrusive and cost-effective method to monitor motion and detect potential fall risks in real time. This project aims to design a system that uses data from wearable devices to predict and prevent falls.

2. Aim:

To develop a wearable sensor-based system that monitors elderly individuals' movements in real time and detects abnormal motion patterns indicative of a fall or high fall risk.

3. Objectives:

- Use wearable sensors such as accelerometers, gyroscopes, and heart rate monitors embedded in smartwatches, belts, or shoe insoles.
- Collect continuous motion data (e.g., gait, posture, acceleration).
- Extract meaningful features like gait velocity, step frequency, body tilt, and sudden shifts in balance.
- Implement machine learning or threshold-based algorithms to detect falls or fall-prone behavior.
- Automatically send alerts to caregivers or medical staff in case of a detected fall or high-risk pattern.
- Provide real-time haptic/audio feedback to the wearer to correct posture or stabilize gait.

4.Expected Outcomes:

- A functional prototype with wearable hardware and fall detection software.
- Accurate identification of high fall-risk situations with low false positives.
- Real-time alert mechanism through mobile app or IoT communication.
- Improved safety and independence for elderly users.
- Dataset of annotated motion data for future research and model refinement.

5.Conclusion:

This project demonstrates how wearable sensor technology can be harnessed to improve elderly care by preventing falls through early detection and real-time intervention. It blends biomedical sensing, machine learning, and IoT to offer a smart health solution that enhances quality of life and reduces hospitalization rates due to falls.

IH3MU23: IoT-based Early Warning and Response System

1. Overview:

This project proposes an IoT-enabled Early Warning and Response System that monitors environmental data in real-time and issues automatic alerts to emergency services and communities.

2. Aim:

To design and prototype an IoT-based system that can detect early signs of natural disasters and initiate timely alerts and response actions to minimize human and material losses.

3. Objectives:

- Install IoT sensors to monitor environmental parameters.
- Use low-power wireless communication protocols to transmit data to a cloud server and implement real-time data logging and visualization.
- Analyze sensor data using rule-based or ML models to detect threshold breaches or patterns indicating potential disasters.
- Send automated alerts via SMS, app notifications, sirens, or public announcement systems.
- Share data and alerts with emergency responders and government dashboards and enable evacuation route suggestions.

4. Expected Outcomes:

- A functional prototype of an IoT-based disaster monitoring system.
- Real-time sensing and alert generation for at least one disaster type (e.g., flood or wildfire).
- A web/mobile dashboard for monitoring sensor status and alerts.
- A communication pipeline for sending alerts to relevant authorities and the public.
- Reduction in response time during disasters due to early detection.

5. Conclusion:

This IoT-based Early Warning and Response System highlights the potential of smart technologies in enhancing disaster preparedness and resilience. By detecting hazards early and automating the response process, this solution can help save lives, reduce damage, and improve coordination between authorities and communities.

IH3MU24: Smart Proctoring System for Exam

1. Overview:

Online examinations have become common in education and recruitment, but maintaining exam integrity remains a major challenge. A Smart Proctoring System uses AI-based tools like facial recognition, object detection, and behavior analysis to monitor examinees in real-time, ensuring fairness and reducing the need for human proctors.

2. Aim:

To develop an AI-driven smart proctoring system that ensures secure, cheat-free online exams through continuous monitoring, authentication, and behavior analysis.

3. Objectives:

- Use facial recognition and ID verification to ensure only authorized candidates can access the exam.
- Monitor candidates via webcam using face detection and gaze tracking
- Capture and analyze audio to detect suspicious conversations or background noises.
- Record screen activity or restrict navigation to prevent opening other windows/apps.
- Automatically log suspicious activities with time-stamped evidence.
- Generate detailed proctoring reports for review.
- Provide an exam dashboard for both candidates and administrators.

4. Expected Outcomes:

- A secure and scalable smart proctoring platform.
- Reduced reliance on manual invigilators.
- Automated alerts for cheating behaviors with evidence.
- Enhanced trust in online examination systems.
- Improved user experience for both students and institutions.

5. Conclusion:

The Smart Proctoring System for Exams integrates AI and computer vision to revolutionize remote assessments. It offers a scalable, accurate, and tamper-resistant solution that can be adopted by educational institutions and certification bodies worldwide, ensuring the integrity and fairness of online exams.

IH3MU25: People Counting Using UAV

1. Overview:

Unmanned Aerial Vehicles (UAVs) offer a flexible and wide-area monitoring solution. By integrating UAVs with AI-based computer vision algorithms, we can automatically count people from aerial images or live video streams with high accuracy.

2. Aim:

To design and implement a UAV-based system that captures aerial data and uses artificial intelligence to detect and count people in real time or from recorded footage.

3. Objectives:

- Use a UAV equipped with a high-definition camera to fly over target areas (e.g., parks, events, city zones).
- Plan optimal flight paths for maximum coverage.
- Capture images or live video footage from various altitudes and angles.
- Apply object detection models (e.g., YOLOv8, SSD, Faster R-CNN) to detect human figures in each frame.
- Use density estimation techniques (e.g., CSRNet, MCNN) for dense crowds where individuals may overlap.
- Present results on a dashboard with counts, density maps, and possible alerts for overcrowding.

4. Expected Outcomes:

- A working prototype that can count people from UAV footage with reasonable accuracy.
- Real-time or near-real-time crowd monitoring system.
- Insights into crowd behavior and movement.
- A dashboard or app to view people count, heatmaps, and flight data.

5. Conclusion:

The People Counting Using UAV project leverages the power of aerial monitoring and artificial intelligence to deliver a scalable, mobile, and efficient crowd analysis tool. This system can be used by law enforcement, event organizers, disaster management authorities, and urban planners to enhance safety, optimize space usage, and respond proactively to emergencies.